

GrounUp Enterprise Specification

Phase 07 - Enterprise Project Management & Operations Platform

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Executive Summary

Phase 07 defines the complete enterprise project management and operations platform for GrounUp. It manages awarded work from project setup and preconstruction through construction, closeout, warranty, and archive. The design unifies project controls, field execution, documents, changes, finances, resources, procurement, safety, quality, mobile/offline work, external collaboration, AI operations, and the Project Digital Twin.

Architecture and Product Model

GrounUp is specified as a multi-tenant SaaS platform with a primary responsive web application, offline-capable mobile clients, external portals, versioned APIs, white-label configuration, and governed AI services. Configuration is preferred over tenant-specific code forks.

Project Command Center

The Command Center is the primary role-aware workspace. It presents project health, schedule, budget, forecast, production, labor, equipment, procurement, safety, quality, documents, pending approvals, alerts, and AI recommendations with drill-down to evidence.

Project Digital Twin

The Digital Twin is a time-stamped, source-aware operational representation assembled from authoritative project records. It displays freshness, confidence, provenance, history, and scenarios while preserving source-system ownership and auditability.

Scheduling and Planning

The platform supports CPM calculations, calendars, logic, constraints, milestones, resource loading, approved baselines, status updates, critical-path analysis, look-ahead planning, pull planning, constraint management, and recovery scenarios.

Field Operations

Field workflows support daily reports, labor, equipment, fuel, production, deliveries, visitors, weather, photos, GPS, voice notes, incidents, inspections, and offline synchronization. Approved field facts update project cost, production, schedule, safety, quality, and the Digital Twin.

Documents, RFIs, and Submittals

Document control preserves immutable revisions, numbering, status, purpose of issue, distribution, acknowledgements, transmittals, search, and supersession. RFIs and submittals use governed review cycles and impact assessment.

Change and Financial Controls

Change events connect scope, estimate, schedule, budget, contract, risk, and forecast effects. Financial controls cover original and revised budget, commitments, actuals, invoices, pay applications, retainage, cash flow, ETC, EAC, variance, and profitability.

Safety, Quality, and Closeout

Safety and quality modules manage plans, JHAs, talks, observations, incidents, inspections, checklists, deficiencies, corrective actions, punch items, closeout deliverables, acceptance, and warranty obligations.

Mobile, External Portals, and White Label

Mobile users receive secure, expiring, project-scoped offline packages. External portals expose only explicitly shared records. Tenants can configure branding, domains, terminology, forms, dashboards, reports, templates, workflows, approvals, custom fields, and enabled modules.

AI Operations

Operational agents may analyze, summarize, forecast, detect anomalies, compare options, and prepare controlled actions. They may not silently approve or post material decisions. Every material output includes evidence, assumptions, confidence, affected records, and version information.

Governance and Release

Phase 07 remains Review-ready until all approval gates are complete. Locking requires passed tests, reconciliations, security review, data-governance review, UAT approval, completed approval record, finalized manifest, and release tag GES-P07-v1.0.0.

Controlled Module Registry

Code	Module	Purpose
PLM	Project Lifecycle	Opportunity through warranty and archive
CMD	Project Command Center	Unified project health, alerts, approvals, and actions
SCH	Scheduling and Planning	CPM, Gantt, look-ahead, pull planning, baselines, constraints
FLD	Field Operations	Daily reports, labor, equipment, production, weather, photos, deliveries
DOC	Document Control	Drawings, specifications, RFIs, submittals, correspondence, revisions
CHG	Change Management	PCOs, directives, change events, pricing, approvals, contract modifications
FIN	Project Financial Controls	Budget, commitments, actuals, forecasts, billing, retainage, cash flow
RES	Resource Management	Crews, labor, equipment, trucks, materials, rentals, subcontractors
QLT	Quality Management	Inspections, checklists, deficiencies, corrective actions, punch lists
SAF	Safety Management	JHAs, toolbox talks, incidents, near misses, inspections, corrective actions
PRC	Procurement and Commitments	Buyout, requisitions, POs, subcontracts, deliveries, lead times
COM	Communications and Meetings	Meeting minutes, action items, notices, correspondence, stakeholder log
DGT	Project Digital Twin	Continuously synchronized operational representation of project
MOB	Mobile and Offline	Role-based field app, offline capture, conflict resolution, secure sync
AI	AI Operations Center	Forecasting, anomaly detection, recommendations, summaries, governed actions
EXT	External Portals	Owner, customer, vendor, subcontractor, inspector, and partner collaboration

Workflow Registry

ID	Workflow	Summary
P07-WF-001	Project Award to Setup	Awarded estimate -> project shell -> contract setup -> WBS/cost codes -> budget import -> team assignment -> kickoff approval
P07-WF-002	Preconstruction Gate	Scope validation -> permits -> procurement plan -> baseline schedule -> safety/quality plans -> mobilization approval
P07-WF-003	Schedule Baseline and Update	Build schedule -> logic review -> resource review -> baseline approval -> periodic status -> variance analysis -> recovery action
P07-WF-004	Daily Field Reporting	Offline/online capture -> validation -> supervisor review -> approval -> cost/production posting -> digital twin refresh
P07-WF-005	RFI Management	Draft -> internal review -> issue -> response -> impact review -> close or route to change event

ID	Workflow	Summary
P07-WF-006	Submittal Management	Register -> prepare -> internal review -> submit -> review cycles -> approved disposition -> distribution
P07-WF-007	Document Revision Control	Receive/create -> classify -> review -> issue -> distribute -> supersede -> acknowledge
P07-WF-008	Change Event to Contract Modification	Identify -> scope review -> estimate -> schedule impact -> negotiate -> approve -> budget/contract update
P07-WF-009	Commitment and Procurement	Need identified -> requisition -> bid comparison -> approval -> PO/subcontract -> delivery/performance -> closeout
P07-WF-010	Invoice and Pay Application	Receive -> validate -> code -> match -> approve -> post -> pay/bill -> retainage update
P07-WF-011	Quality Deficiency	Inspect -> record deficiency -> assign -> correct -> verify -> close -> trend analysis
P07-WF-012	Safety Incident	Report -> immediate response -> notifications -> investigation -> corrective actions -> regulatory/claim follow-up -> closure
P07-WF-013	Risk and Issue Management	Identify -> assess -> assign -> response plan -> monitor -> escalate -> close/convert
P07-WF-014	Project Closeout	Readiness check -> punch -> as-builts/O&M/warranties -> financial close -> acceptance -> archive
P07-WF-015	Offline Synchronization	Prepare scoped package -> encrypt -> capture changes -> reconnect -> validate -> conflict resolve -> post -> audit

Engine Registry

ID	Engine	Purpose
P07-ENG-001	Project Lifecycle Engine	Controls project states, gates, required artifacts, waivers, and transitions.
P07-ENG-002	Command Center Aggregation Engine	Assembles role-specific project health, alerts, approvals, KPIs, and actions.
P07-ENG-003	CPM Scheduling Engine	Calculates early/late dates, float, critical path, calendars, constraints, and baselines.
P07-ENG-004	Look-Ahead and Constraint Engine	Builds rolling plans and tracks constraint readiness.
P07-ENG-005	Resource Leveling Engine	Detects conflicts and recommends labor/equipment/crew allocation options.
P07-ENG-006	Field Capture and Posting Engine	Validates and posts daily labor, equipment, production, weather, and delivery data.
P07-ENG-007	Document Control Engine	Manages numbering, revisions, transmittals, review cycles, acknowledgements, and supersession.
P07-ENG-008	RFI and Submittal Engine	Controls registers, due dates, workflows, impacts, and response cycles.
P07-ENG-009	Change Intelligence Engine	Links scope changes to estimate, budget, schedule, contract, and forecast impacts.
P07-ENG-010	Project Cost Control Engine	Maintains budget, commitments, actuals, forecast, variance, and cost-to-complete.
P07-ENG-011	Billing and Retainage Engine	Calculates progress billing, stored material, retainage, and payment status.
P07-ENG-012	Procurement and Lead-Time Engine	Tracks buyout, procurement status, lead times, required-on-site dates, and delivery risk.
P07-ENG-013	Quality Control Engine	Executes inspection plans, checklists, deficiencies, corrective actions, and punch lists.
P07-ENG-014	Safety Control Engine	Manages plans, JHAs, talks, observations, incidents, and corrective actions.
P07-ENG-015	Digital Twin Synchronization Engine	Creates time-stamped, source-aware operational snapshots and freshness indicators.

ID	Engine	Purpose
P07-ENG-016	Project Risk Scoring Engine	Scores cost, schedule, safety, quality, procurement, and documentation risk.
P07-ENG-017	Forecast and Scenario Engine	Produces EAC, ETC, cash flow, completion, and recovery scenarios.
P07-ENG-018	Mobile Offline Sync Engine	Scopes, encrypts, validates, reconciles, and audits offline data synchronization.
P07-ENG-019	Notification and Escalation Engine	Applies subscriptions, severity, SLA, digest, escalation, and acknowledgement logic.
P07-ENG-020	Project Audit and Reconciliation Engine	Detects cross-module mismatches and maintains immutable transaction evidence.

Business Rules

ID	Rule
P07-BR-001	Project creation requires an owning tenant, company, project manager, base currency, timezone, and lifecycle template.
P07-BR-002	A project may not enter Construction until required preconstruction gates are approved or formally waived.
P07-BR-003	Approved estimate values become the initial control budget through a traceable import, not direct copy-over.
P07-BR-004	Original budget, contract value, and approved baselines are immutable; adjustments use controlled revisions.
P07-BR-005	Only one active approved schedule baseline may exist per baseline type and project.
P07-BR-006	Progress updates must use a defined data date and retain prior schedule-update snapshots.
P07-BR-007	Daily reports may be edited until submitted; post-approval corrections require a revision record.
P07-BR-008	Labor, equipment, and production entries must map to authorized cost codes and project locations when required.
P07-BR-009	Document revisions never overwrite prior revisions and must clearly identify the current revision.
P07-BR-010	RFI and submittal responses cannot be treated as contract changes unless routed through change management.
P07-BR-011	Change events must separately identify scope, cost, schedule, risk, and contractual impacts.
P07-BR-012	Committed cost must originate from an approved commitment or approved controlled exception.
P07-BR-013	Forecast-at-completion must reconcile approved budget, commitments, actual costs, pending changes, and remaining forecast.
P07-BR-014	External users may access only explicitly shared records and redacted fields within an active access period.
P07-BR-015	Offline changes require conflict checks against server versions before writeback.
P07-BR-016	AI recommendations remain advisory until accepted by an authorized user; high-impact actions require approval.
P07-BR-017	Digital twin snapshots must preserve source timestamps and data freshness indicators.
P07-BR-018	Safety incidents and serious quality events require immediate escalation under tenant-defined thresholds.
P07-BR-019	Project closeout requires completion or approved waiver of defined closeout deliverables.
P07-BR-020	All material actions and approvals must create immutable audit events.

Validation Rules

ID	Rule
P07-VR-001	Project dates must be logically ordered and use the project timezone.
P07-VR-002	Percent complete must be between 0 and 100 and conform to the selected progress method.
P07-VR-003	Schedule relationships may not create unapproved circular logic.
P07-VR-004	Actual finish requires actual start and 100 percent completion.
P07-VR-005	Budget and cost transaction amounts require currency and non-null cost-code mapping.

ID	Rule
P07-VR-006	Daily labor hours exceeding tenant thresholds require explanation and approval.
P07-VR-007	Equipment meter readings may not decrease unless a documented meter replacement or correction exists.
P07-VR-008	Production quantities require a valid unit and cannot duplicate the same source event.
P07-VR-009	Document revision identifiers must be unique within the document series.
P07-VR-010	RFI due dates may not precede the issue date.
P07-VR-011	Submittal review cycles must preserve reviewer, disposition, dates, and returned files.
P07-VR-012	Change-order values must equal the sum of approved change lines within tolerance.
P07-VR-013	Invoices may not exceed commitment balance without a controlled exception.
P07-VR-014	Retainage percentages must remain within contract-defined limits.
P07-VR-015	Inspection results require checklist completion or documented not-applicable reasons.
P07-VR-016	Incident records require severity, date/time, location, reporter, and notification status.
P07-VR-017	Offline payloads must pass schema, authorization, version, and malware checks.
P07-VR-018	External portal shares require an expiration date unless policy explicitly allows otherwise.
P07-VR-019	AI-generated claims must include confidence, evidence references, and model/task version.
P07-VR-020	Closeout cannot be approved while mandatory unresolved punch or warranty-prep items remain.

KPI Registry

ID	KPI	Definition	Unit
P07-KPI-001	Project Health Score	Weighted score of cost, schedule, safety, quality, procurement, document, and risk indicators	0-100
P07-KPI-002	Cost Performance Index	Earned value divided by actual cost	Ratio
P07-KPI-003	Schedule Performance Index	Earned value divided by planned value	Ratio
P07-KPI-004	Forecast Variance	Approved budget minus forecast at completion	USD
P07-KPI-005	Committed Cost Coverage	Commitments divided by forecasted buyout scope	Percent
P07-KPI-006	Change Exposure	Pending and unapproved change value	USD
P07-KPI-007	Critical Activity Slippage	Critical-path delay versus approved baseline	Days
P07-KPI-008	Constraint Readiness	Ready look-ahead activities divided by planned activities	Percent
P07-KPI-009	Labor Productivity Variance	Actual production rate versus planned production rate	Percent
P07-KPI-010	Equipment Utilization	Productive equipment hours divided by available hours	Percent
P07-KPI-011	RFI Aging	Average open days for RFIs	Days
P07-KPI-012	Submittal On-Time Rate	Submittals completed on or before required date	Percent
P07-KPI-013	Procurement On-Time Rate	Items delivered by required-on-site date	Percent
P07-KPI-014	Rework Cost Rate	Rework cost divided by total installed cost	Percent
P07-KPI-015	Inspection Pass Rate	Passed inspections divided by completed inspections	Percent
P07-KPI-016	Punch Closure Velocity	Punch items closed per week	Count/week
P07-KPI-017	Recordable Incident Rate	Tenant-defined normalized recordable incident rate	Rate
P07-KPI-018	Daily Report Timeliness	Daily reports approved by cutoff	Percent
P07-KPI-019	Cash Conversion Forecast	Forecast time from earned revenue to collected cash	Days
P07-KPI-020	Closeout Readiness	Completed mandatory closeout requirements	Percent